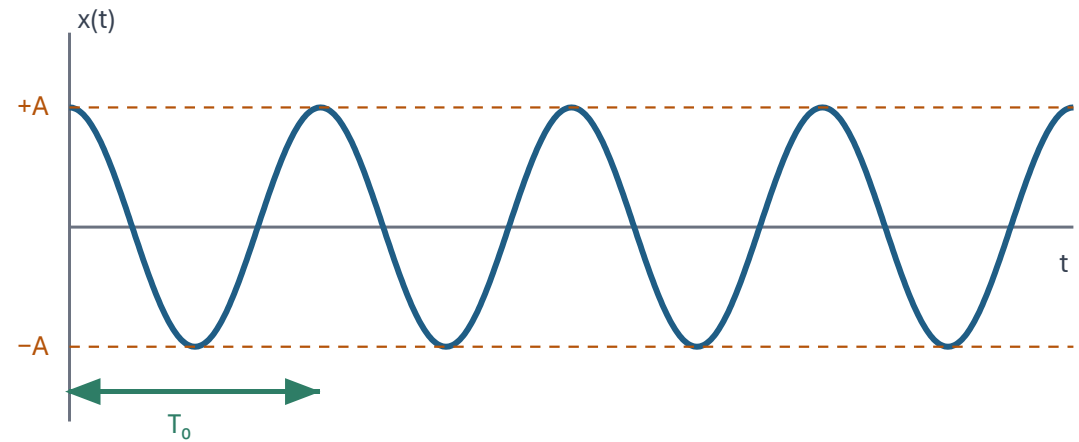


## Topic 2

# Sinusoids

From cosine waves to complex phasors



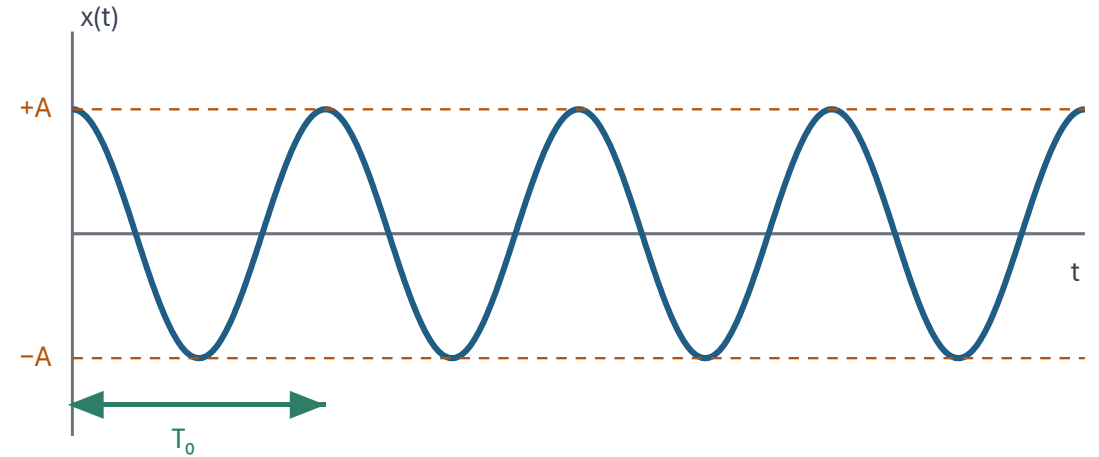
**Key idea.** Sinusoids are the basic oscillating building blocks of signals. Complex exponentials turn trigonometric manipulations into vector algebra.

# What is a sinusoid?

*A cosine with a linearly increasing phase*

$$x(t) = A\cos(\omega_0 t + \phi)$$

- $A$ : amplitude / peak value
- $\omega_0$ : radian frequency in rad/s
- $\phi$ : initial phase in radians
- The signal repeats with period  $T_0$

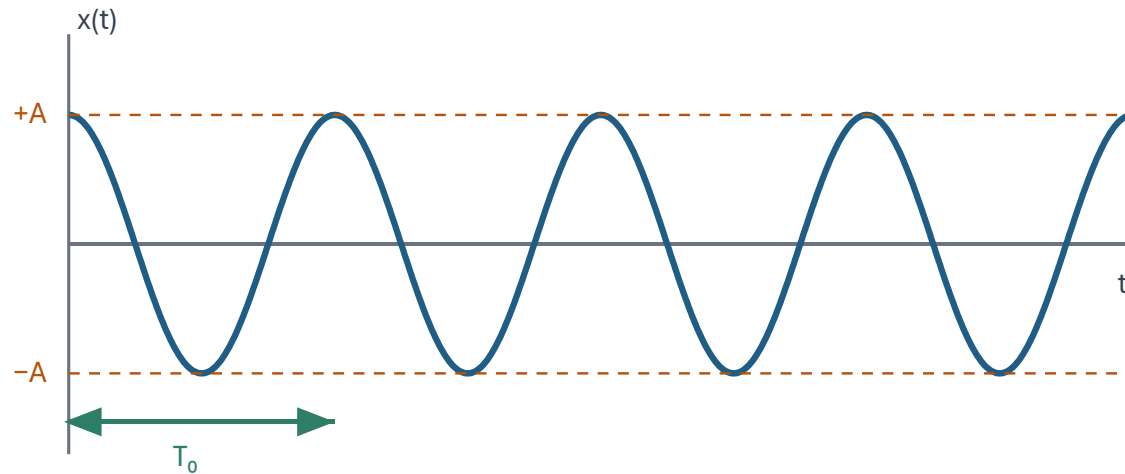


$$f_0 = \frac{\omega_0}{2\pi}, \quad T_0 = \frac{1}{f_0} = \frac{2\pi}{\omega_0}$$

$A$  changes vertical scale;  $\omega_0$  changes how fast it oscillates;  $\phi$  shifts the waveform horizontally.

# Physical motivation: the A-440 tuning fork

*A near-pure tone is well modelled by a sinusoid*



one cycle every 2.27 ms

$$x(t) = 10\cos(2\pi \cdot 440 \cdot t - 0.4\pi)$$

- Frequency: 440 Hz — A above middle C.
- Period:  $1/440$  s  $\approx$  2.27 ms.
- Amplitude controls loudness-related scale, not pitch.
- Initial phase sets where the cycle starts at  $t=0$ .

# Sine and cosine toolbox

*Identities used throughout sinusoidal analysis*

|                      |                                                                                   |
|----------------------|-----------------------------------------------------------------------------------|
| Phase relation       | $\sin(\theta) = \cos(\theta - \pi/2), \quad \cos(\theta) = \sin(\theta + \pi/2)$  |
| Symmetry             | $\cos(-\theta) = \cos(\theta), \quad \sin(-\theta) = -\sin(\theta)$               |
| Pythagorean identity | $\sin^2(\theta) + \cos^2(\theta) = 1$                                             |
| Angle sum            | $\cos(\alpha \pm \beta) = \cos\alpha \cos\beta \mp \sin\alpha \sin\beta$          |
| Derivatives          | $d \sin\theta / d\theta = \cos\theta, \quad d \cos\theta / d\theta = -\sin\theta$ |

*Use these identities to move between sine/cosine forms and to derive phasor relationships.*

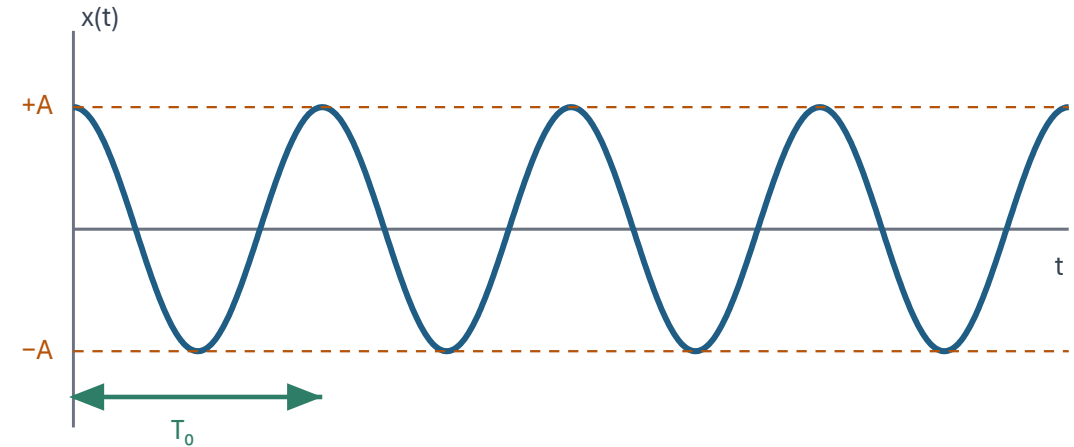
# Frequency and period

Two equivalent ways to describe how fast a sinusoid oscillates

$$x(t + T_0) = x(t)$$

$$\omega_0 T_0 = 2\pi \quad \Rightarrow \quad T_0 = \frac{2\pi}{\omega_0}$$

$$f_0 = \frac{1}{T_0}$$



## Example

A 440 Hz sinusoid completes 440 cycles each second. The period is approximately 2.27 ms.

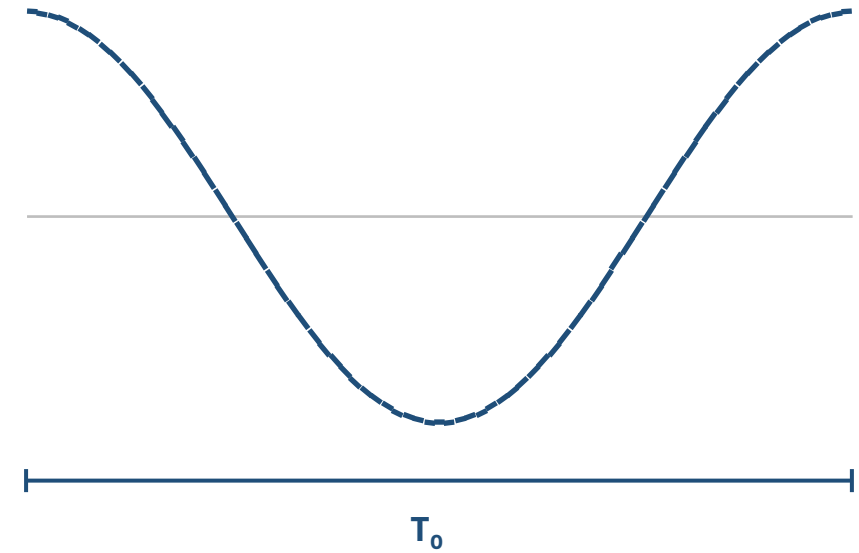
# Example 1: frequency and period

Read sinusoidal parameters from the formula

$$x(t) = 2 \cos(0.4\pi t)$$

Find  $A$ ,  $\omega_0$ ,  $f_0$ , and  $T_0$ .

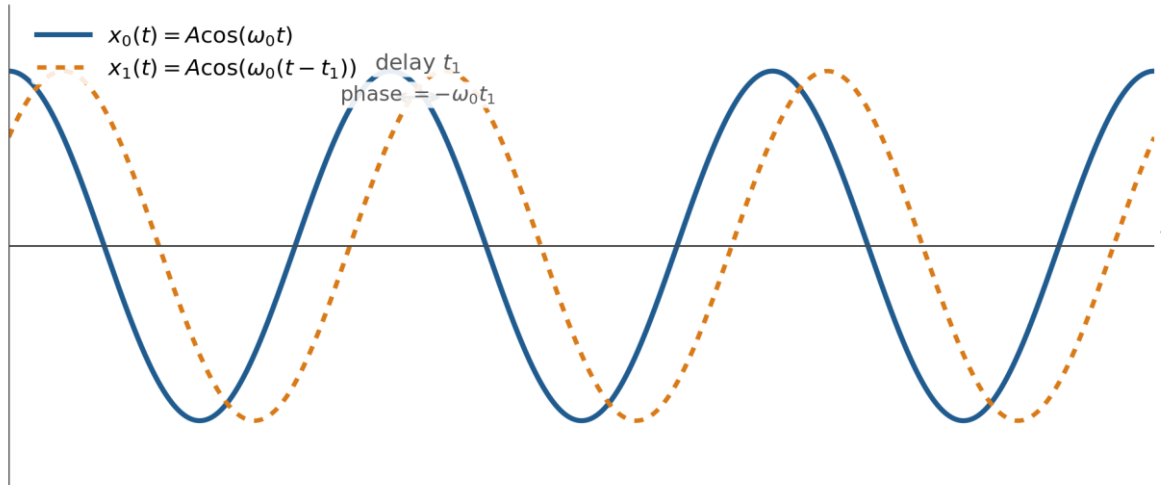
- 1 Amplitude is the coefficient of the cosine:  $A = 2$ .
- 2 Compare with  $A \cos(\omega_0 t + \phi)$ :  $\omega_0 = 0.4\pi$  rad/s.
- 3 Convert to cyclic frequency:  $f_0 = \omega_0/(2\pi) = 0.2$  Hz.
- 4 Period:  $T_0 = 1/f_0 = 5$  s.



one cycle = 5 s

# Phase shift and time shift

*A time delay appears as a negative phase shift*



$$A \cos(\omega_0(t - t_1)) = A \cos(\omega_0 t - \omega_0 t_1)$$

- If  $t_1 > 0$ : the waveform moves right; it is delayed.
- The corresponding phase is  $\phi = -\omega_0 t_1$ .
- Phase is ambiguous modulo  $2\pi$ .
- A common convention is  $-\pi < \phi \leq \pi$ .

$$t_1 = -\frac{\phi}{\omega_0}$$

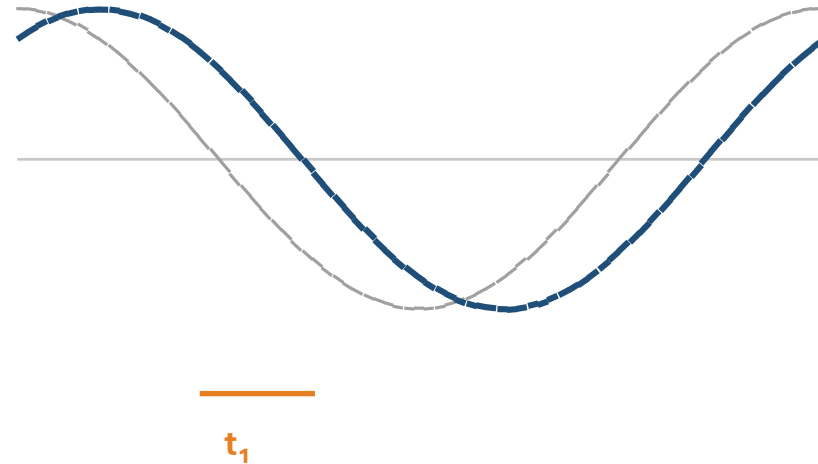
## Example 2: phase and delay

A negative phase corresponds to a positive delay

$$x(t) = 3 \cos(20\pi t - 0.2\pi)$$

**Rewrite it as  $A \cos[\omega_0(t - t_1)]$ .**

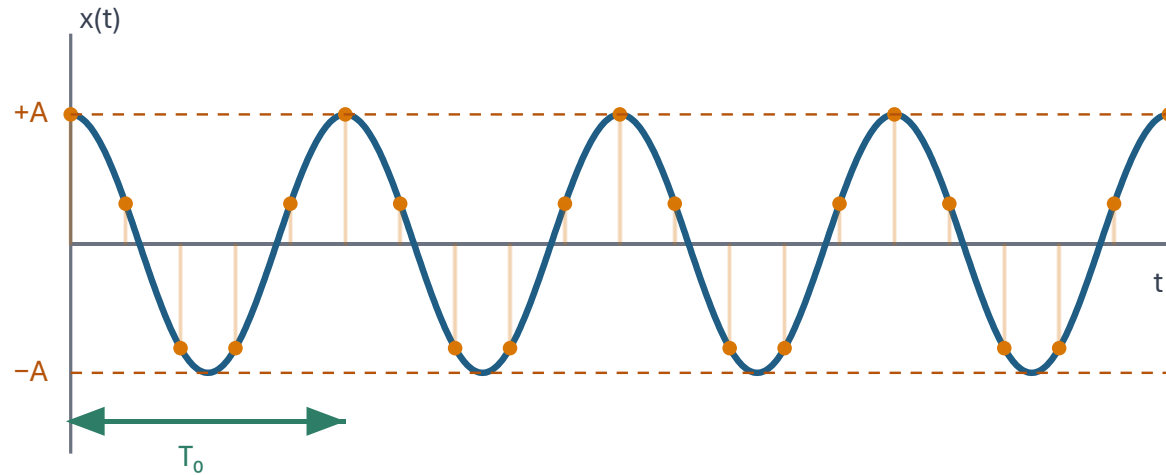
- 1 Identify  $\omega_0 = 20\pi$  rad/s and  $\phi = -0.2\pi$ .
- 2 Use  $\phi = -\omega_0 t_1$ , so  $t_1 = -\phi/\omega_0$ .
- 3  $t_1 = 0.2\pi / 20\pi = 0.01$  s = 10 ms.
- 4 Therefore  $x(t) = 3 \cos[20\pi(t - 0.01)]$ .



delay:  $t_1 = 10$  ms

# Sampling and plotting sinusoids

A continuous-time sinusoid becomes a sequence evaluated at  $t_n = nT_s$



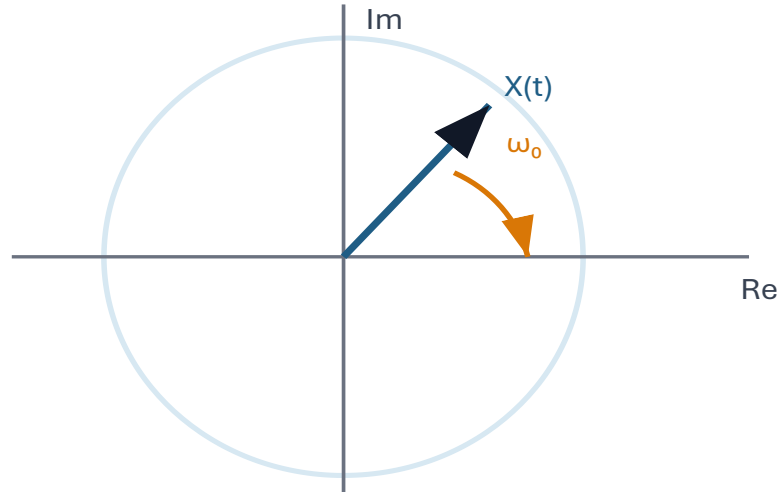
Smaller  $T_s \Rightarrow$  more samples per cycle  $\Rightarrow$  smoother plot and better time resolution.

$$x[n] = x(nT_s) = A\cos(\omega_0 nT_s + \phi)$$

```
n = np.arange(-7, 6)
Ts = 0.005
tn = n * Ts
xn = 20*np.cos(80*np.pi*tn - 0.4*np.pi)
plt.plot(tn, xn)
```

# Complex numbers as vectors

*The polar form makes rotations and scaling transparent*



$$z = x + jy = re^{j\theta}$$

- Magnitude  $r$  is the vector length.
- Angle  $\theta$  is measured from the real axis.
- Multiplication multiplies magnitudes and adds angles.
- This is why complex numbers are useful for sinusoids.

$$r = |z| = \sqrt{x^2 + y^2}, \quad \theta = \text{atan2}(y, x)$$

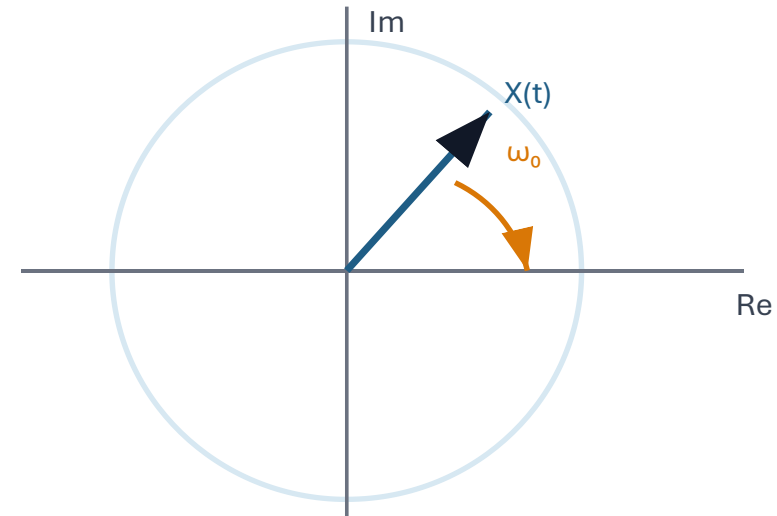
# Euler's formula

*The bridge between trigonometry and complex exponentials*

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$



Interpretation:  $e^{j\theta}$  is a unit-length vector at angle  $\theta$  on the complex plane.

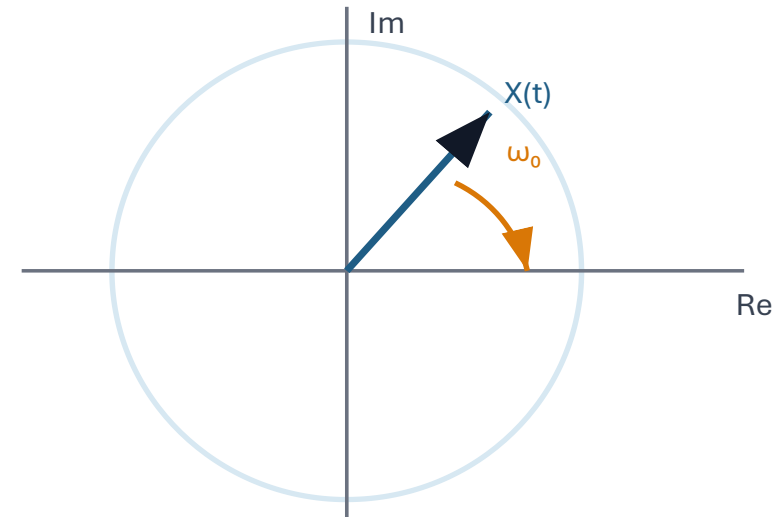
# Complex exponential signals

*A rotating complex number*

$$x_c(t) = X e^{j\omega_0 t}$$

$$X = A e^{j\phi}$$

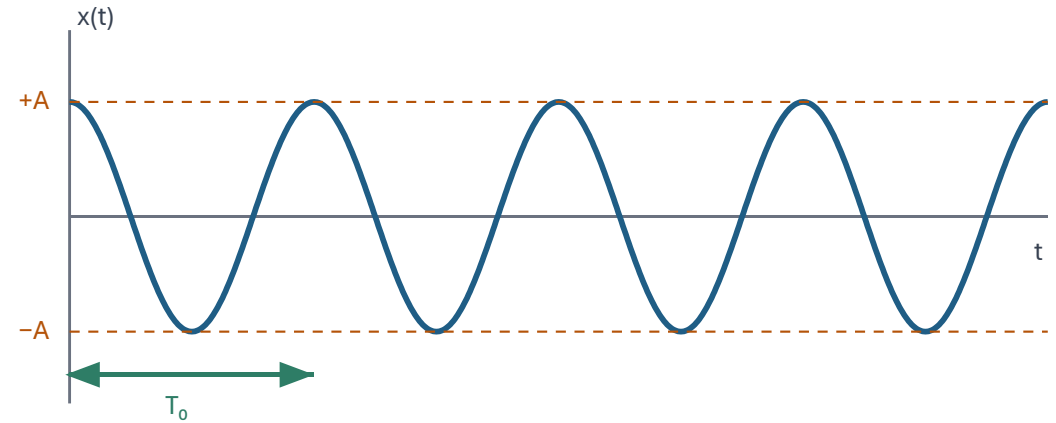
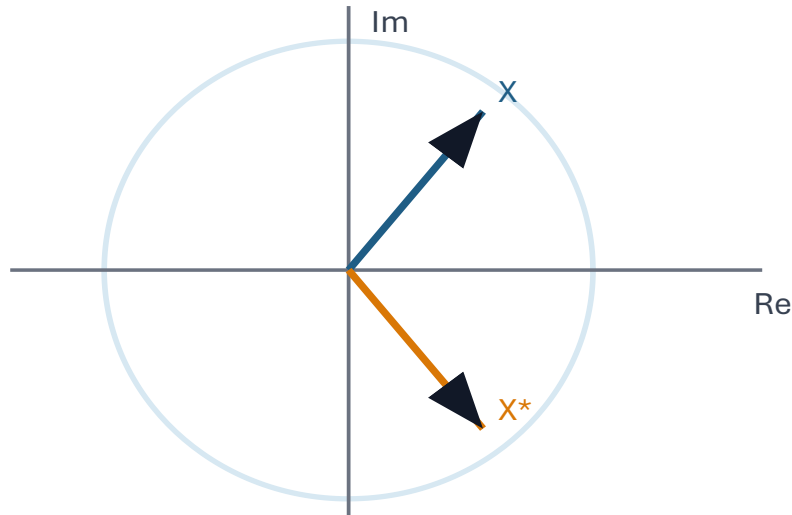
- $X$  is the complex amplitude or phasor.
- $e^{j\omega_0 t}$  rotates the phasor with angular speed  $\omega_0$ .
- Positive  $\omega_0$  rotates counter-clockwise.
- One full revolution corresponds to  $2\pi$  radians.



# A real cosine as two rotating phasors

*Positive and negative frequencies appear as conjugate partners*

$$A\cos(\omega_0 t + \phi) = \frac{1}{2}Ae^{j\phi}e^{j\omega_0 t} + \frac{1}{2}Ae^{-j\phi}e^{-j\omega_0 t}$$



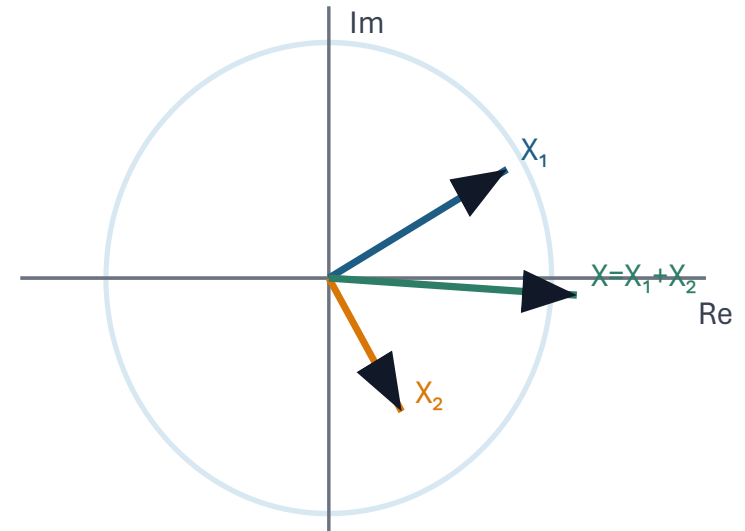
- The conjugate pair makes the sum real.
- One phasor rotates at  $+\omega_0$ ; the other at  $-\omega_0$ .
- The real part is the observed cosine.

# Phasor addition

Same-frequency sinusoids can be combined by adding complex numbers

$$\sum_k A_k \cos(\omega_0 t + \phi_k) = A \cos(\omega_0 t + \phi)$$

- Convert each sinusoid to a phasor  $A_k e^{j\phi_k}$ .
- Add the phasors as complex numbers.
- Convert the resulting phasor  $A e^{j\phi}$  back to cosine form.



$$A e^{j\phi} = \sum_k A_k e^{j\phi_k}$$

## Example 3: adding two sinusoids

Use the phasor method instead of trigonometric expansion

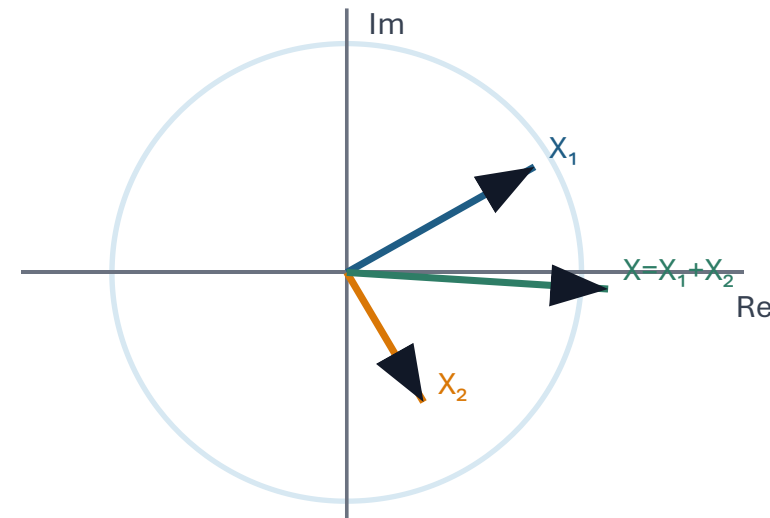
$$x(t) = 3\cos(\omega t + \pi/6) + 2\cos(\omega t - \pi/3)$$

### Phasors

$$X_1 = 3e^{j\pi/6}, \quad X_2 = 2e^{-j\pi/3}$$

$$X = X_1 + X_2 \approx 3.598 - j0.232$$

$$|X| \approx 3.61, \quad \angle X \approx -0.065 \text{ rad}$$



$$x(t) \approx 3.61\cos(\omega t - 0.065)$$

# Notation and takeaways

*The minimum language for working with sinusoids*

| Symbol     | Meaning                         |
|------------|---------------------------------|
| $A$        | Amplitude / peak value          |
| $\omega_0$ | Radian frequency, rad/s         |
| $f_0$      | Frequency in hertz, Hz          |
| $\phi$     | Initial phase angle             |
| $T_0$      | Fundamental period, $T_0=1/f_0$ |
| $T_s$      | Sampling period                 |
| $X$        | Complex amplitude or phasor     |

## Three core takeaways

- A sinusoid is fully described by amplitude, frequency, and phase.
- Sampling evaluates the continuous waveform at  $t = nT_s$ .
- Complex phasors make sinusoidal algebra graphical and linear.

sinusoids + phasors = simpler signal analysis